

UNIT 3

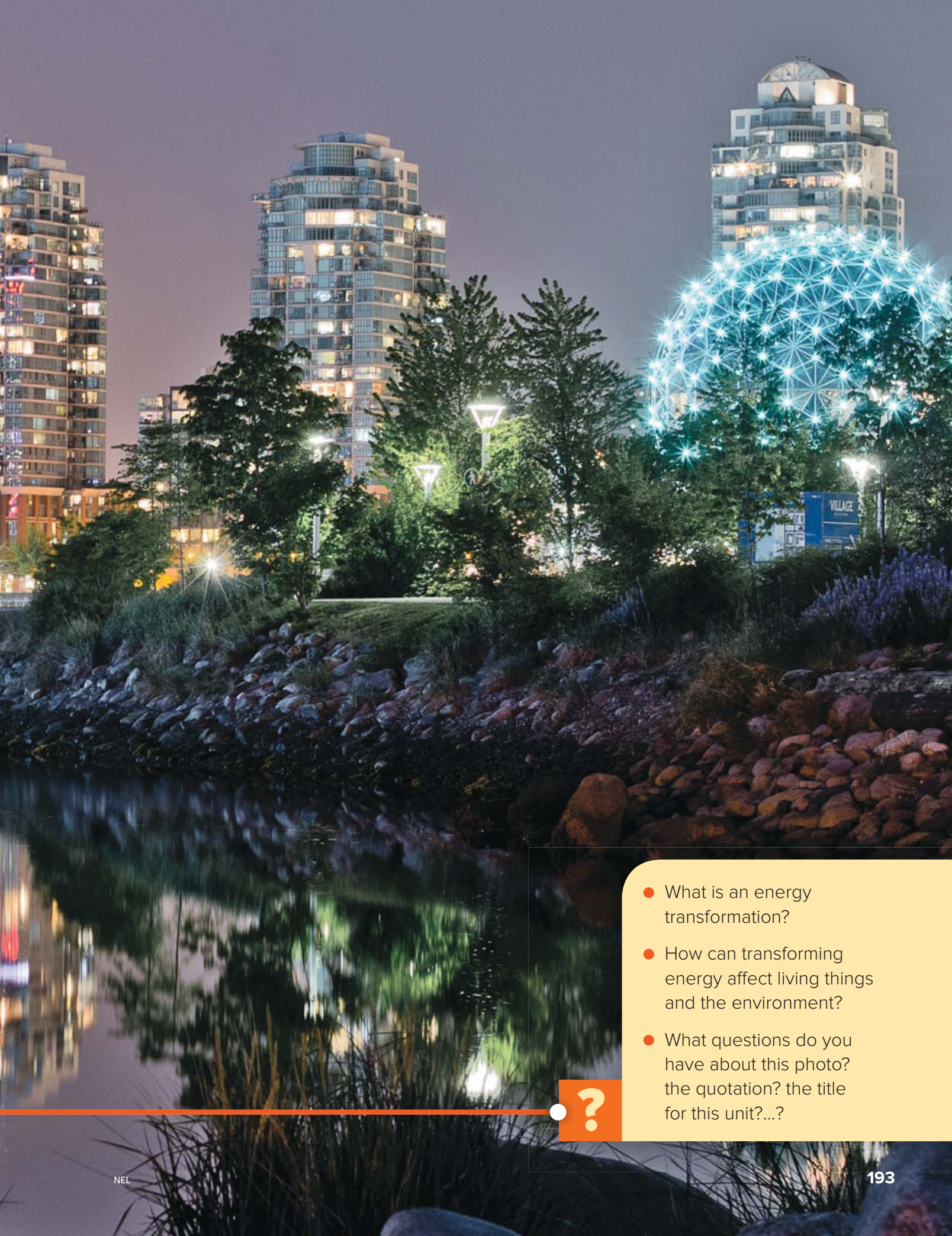
Energy is conserved and its transformation affects living things and the environment.

It takes an enormous amount of electrical energy to light up a city. This energy is transformed from other types of energy—the energy stored in fossil and nuclear fuels, and the energy of the Sun, moving water, and wind. In the future, buildings and the materials used to make them may play a role in transforming energy as well.

“ In the cities of the future, advances such as glass that can harness the sun’s power—so skyscrapers could be covered in windows that double as solar panels—have the potential to propel widespread use of solar energy. ”

*Dr. James Tansey,
executive director of the UBC Sustainability
Initiative and the Sauder Centre for
Social Innovation and Impact Investing,
University of British Columbia*





- What is an energy transformation?
- How can transforming energy affect living things and the environment?
- What questions do you have about this photo? the quotation? the title for this unit?...?



At a Glance

You will demonstrate what you know, can do, and understand by being able to

- Design and perform investigations to explore the nature and effects of energy transfer and transformation
- Develop and use models and other methods to show connections between different forms of energy
- Develop evidence-based explanations about beneficial and harmful applications of energy transfer and transformation
- Seek patterns and connections to describe, explain, and evaluate local and global impacts of energy transformations

TOPIC 3.1:

What are the properties of energy?

Some things you will do:

- seek and analyze patterns, trends, and connections in data
- analyze cause-and-effect relationships

Some things you will come to know:

- We can investigate energy by observing its effects on other things.
- There are different forms of energy, which can be classified under the categories of kinetic and potential.
- When energy is transferred or transformed, the total amount of energy involved is conserved.

ESSENTIAL QUESTION

How can we investigate and apply our understanding of energy transfer and transformation?





TOPIC 3.2:

How is energy transformed?

Some things you will do:

- critically analyze the validity of information and evaluate approaches used to solve problems
- generate and introduce new or refined ideas when problem solving

Some things you will come to know:

- Energy is transformed in chemical reactions as well as in nuclear reactions.
- Life as we know it could not exist without energy transformations associated with photosynthesis and cellular respiration.
- Nuclear fission and nuclear fusion are the two types of nuclear reactions.



TOPIC 3.3:

How does energy transformation affect global systems?

Some things you will do:

- use knowledge of scientific concepts to draw conclusions that are consistent with evidence
- demonstrate an awareness of assumptions and bias in your own work and in secondary sources

Some things you will come to know:

- The transformation and transfer of energy have local, national, and global effects on weather and climate.
- Negative consequences of energy transformation and transfer must be understood in order to be addressed.



TOPIC 3.4:

How does energy transformation affect humans?

Some things you will do:

- contribute to finding solutions to problems at a local and/or global level through inquiry
- consider changes in knowledge over time as tools and technologies have developed

Some things you will come to know:

- We depend on First Peoples and other ways of knowing to help us revise, extend, and complement ideas of Western science.
- We have responsibilities as individuals, members of local communities, and members of the global community to think and act responsibly with regard to energy-related knowledge and technologies.

Connect To What You Already Know

This feature helps you reflect on what you know about some foundational ideas that you have learned in previous grades. Work alone or quietly in small groups to answer the questions. Reach out to your classmates to ask about things that you are unsure about or to offer assistance. Your teacher also can provide additional reinforcement materials to help you prepare for this unit.

1. Compare and contrast the wave model of light and the particle model of light.
2. Use **Figure 1** on the next page to create a table comparing types of electromagnetic radiation in terms of wavelength, frequency, uses, and any other properties you can think of.
3. Describe three examples of how you interacted with different types of electromagnetic radiation today.
4. Different observations and experiments have led to an understanding of the various properties of electromagnetic radiation. Describe at least three of these properties.
5. Electromagnetic radiation is released by the unstable nuclei of certain atoms.
 - a) What are atoms with unstable nuclei called?
 - b) What type or types of electromagnetic radiation do these nuclei release?
 - c) Is this electromagnetic radiation harmful to living things? Explain your answer.

6. The incredible amount of energy generated on and emitted by the Sun supports life on Earth. Use a labelled sketch to explain the relationships among nuclear fusion, solar energy, and electromagnetic radiation.
7. Use **Figure 2** on the next page to help you describe the role of greenhouse gases in maintaining Earth's average global temperature.
8. Use a Venn diagram to compare and contrast global warming and global climate change.
9. Copy and complete the following table.

Greenhouse Gas	Natural Sources	Human Sources
carbon dioxide		
methane		
nitrous oxide		
water vapour		

10. Describe three ways that Earth's systems may be affected by an increase in greenhouse gases.
11. Agree or disagree with the following statements. Give reasons for your views.
 - a) Global climate change is a consequence of increased levels of greenhouse gases in the atmosphere.
 - b) Global climate change will affect Earth's temperature, but not precipitation amounts, wind patterns, or storm severity.
12. Think about what you know about the carbon cycle. Use a flowchart to show
 - a) how carbon moves between the biosphere, geosphere, hydrosphere, and atmosphere when fossil fuels (coal, oil, and natural gas) are formed, extracted, and burned
 - b) how solar energy travels through and interacts with Earth's spheres

13. Create a comic strip or infographic that shows three ways that humans could act to reduce the amount of greenhouse gases that we contribute to the atmosphere.

14. You have answered questions about several different topics. Use a graphic organizer to identify these topics and outline your understanding of them.

Figure 1 The electromagnetic spectrum is a model that shows the range (spectrum) of electromagnetic radiation in terms of wavelength.

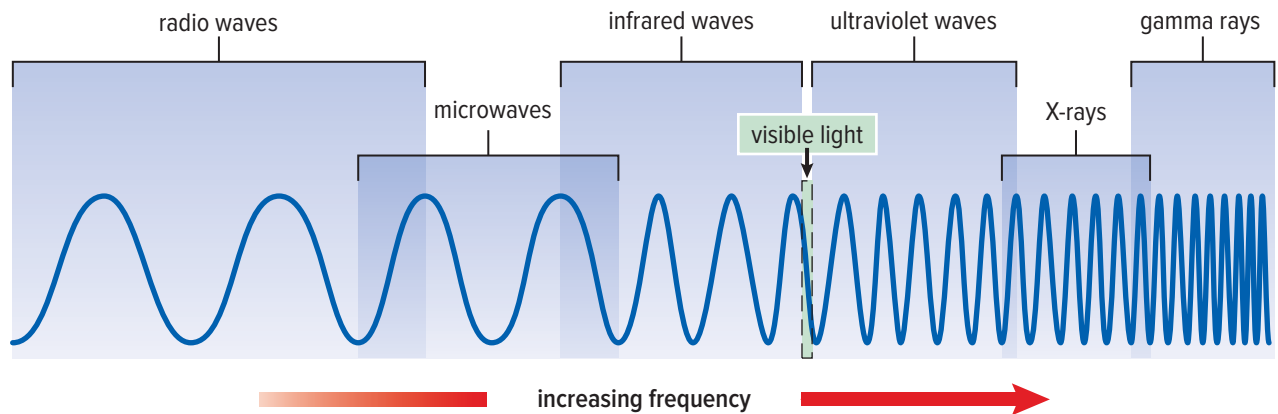


Figure 2 Most of the solar energy that enters Earth's atmosphere is absorbed by land and water at the surface. Some is reflected by the surface and the atmosphere and passes back into space. Some of the absorbed energy is re-emitted as infrared radiation and absorbed by gases in the atmosphere. The gases that absorb this energy are called greenhouse gases.

